

Evidence Room

Agents that earn responsibility.

AP Agent OS Pro — Operating System Overview

Governance spine for an AP agent workforce — beside your stack, not instead of it.

SKU ER-PRO · Operating system overview (not full agent dump) · 20–28 pages

v1.0 · Commercial pack · Not legal advice · Not a software runtime

1. What an AP Agent OS is

AP Agent OS Pro is the governed layer between your ERP/AP automation stack and AI agents that assist invoice-to-pay. It supplies process maps, agent role cards, promotion gates, controls, KPI measurement, illustrative business-case worksheets, templates, and testing packs.

It does not replace Coupa, Tipalti, SAP, Oracle, or your auditor. Think of it as the missing operating system for an agent workforce: charters, evidence standards, earn-up, and demotion — the same discipline you would apply to a new hire, applied to software that can draft and (eventually, narrowly) act.

2. Design principles

- Evidence before autonomy.
- Promotion gates for agents; accountability for humans.
- Beside the stack — not a rip-and-replace fantasy.
- Propose-only for payment; dual control for bank changes.
- Illustrative models labeled ILLUSTRATIVE; benchmarks labeled EXTERNAL.
- Demotion is hygiene, not theatre.

3. Logical architecture

- **Process Mapping** — current-state invoice-to-pay lifecycle and exception taxonomy.
- **Agent Library** — A01–A16 role cards with stage paths and forbidden actions.
- **Governance** — earn-up gates, RACI, kill-switches, change control for prompts/models.
- **Controls** — preventive and detective controls mapped to agent risks.
- **KPI Measurement** — activity, operational, financial, and risk-control metrics.
- **Business Case** — illustrative models; your inputs required.
- **Templates & Testing** — charters, evidence packs, pilot test scripts.

4. Earn-up as the control plane

Observe → Recommend → Draft → Execute (narrow) → Expand. Each promotion requires sampled evidence, owner sign-off, and intact kill-switches. Demotion is first-class when precision fails.

Stage	Agent may	Agent may not	Gate to next
Observe	Read signals; log	Outbound action	Coverage + noise review
Recommend	Rank options	Decide	Precision sample + owner accept
Draft	Prepare artefacts	Send/post alone	Edit-distance / QA sample
Execute (narrow)	Act on whitelist classes	Act outside class / money release	Earn-in metrics + Controller sign-off
Expand	Widen classes carefully	Skip sampling	Board roadmap + capacity

5. Agent portfolio map (overview — not full dump)

This overview intentionally omits full playbook text. Pro ZIP packs contain the complete library.

Band	Agents	Primary risk focus
Intake & validate	A01–A02	Garbage-in; master-data damage
Match & triage	A03–A04	False match; wrong routing
Upstream quality	A05–A06	GR/PO defects flooding AP
Decision support	A07–A09	SoD; supplier/internal chase quality
Signals	A10	Duplicate/anomaly — signals only
Close & stewardship	A11–A14	Statement, proposal, close, report drafts
Improve & orchestrate	A15–A16	Root cause; queue orchestration

6. Band narratives — Intake through Signals

Intake & validate (A01–A02)

Intake creates the Invoice Case. Validation fails closed on structural and master-data defects before matching. Pilot on one channel; measure capture completeness and false holds.

Match & triage (A03–A04)

Matching proposes dispositions within published tolerances. Triage assigns taxonomy codes and owners. False auto-match is more expensive than slow human review — precision gates before any bounded execute.

Upstream quality (A05–A06)

Many AP exceptions are born in receiving and procurement. These agents signal; they do not fabricate GR or rewrite POs.

Decision support (A07–A09)

Approval routing must preserve SoD. Supplier and internal follow-up agents draft; humans send until earn-in on approved templates.

Signals (A10)

Duplicate and anomaly agents emit investigation signals. Never market as fraud guarantees. Never auto-void.

7. Band narratives — Close, Improve, Orchestrate

Close & stewardship (A11–A14)

Statement reconciliation, payment proposal review (propose only), close task drafting, and reporting packs — all under human accountability. A12 must never hold release authority.

Improve & orchestrate (A15–A16)

Root-cause analysis drafts and queue orchestration recommendations for the AP Manager — not autonomous reallocation of payment authority.

8. Control themes

- Fail-closed validation for money-adjacent fields
- Propose-only for payment; human release identity required
- Dual control on vendor bank changes — independent of agents
- Version-pinned prompts/models in evidence packs
- Sampled QA with demotion triggers
- Data minimisation toward external LLM tools
- Change control for every prompt/model promotion
- Kill-switch tested before first Propose

9. Controls matrix (illustrative pattern)

Scope	Risk	Control type	Evidence
A01 Intake	Incomplete case enters queue	Preventive gate	Source hash; confidence; edit log
A03 Match	False auto-match	Detective sample	Line compare; policy version
A07 Approval	SoD breach	Preventive matrix	Route path; matrix version
A10 Duplicate	Missed duplicate paid	Detective + human	Signal set; investigator notes
A12 Proposal	Agent releases payment	Preventive forbid	Proposal packet; human release ID
Bank change	Fraudulent remit	Dual control	Call-back log; dual approver IDs
All agents	Silent model change	Change control	Ticket; model hash

10. KPI spine

Track activity (volume, agent-assisted share, evidence packs), operational (STP, exception rate, first-pass match, cycle time), financial (cost/invoice, on-time pay, discount capture), and risk-control (SoD breaches, force-pays, duplicate precision, dual-control %). Definitions before targets; baselines before pilots.

Category	Example KPI	Cadence	Owner
Activity	Agent-assisted volume %	Weekly	Agent Owner
Operational	Exception rate	Weekly	Exception Lead
Operational	First-pass match rate	Weekly	Matching Lead
Financial	Cost per invoice	Quarterly	Controller
Financial	On-time payment rate	Monthly	Treasury/AP
Risk-control	SoD violations	Monthly	Internal Controls
Risk-control	Duplicate suspect precision	Monthly	A10 Owner

11. Measurement hygiene

- Write definitions before the first dashboard.
- Document exclusions (intercompany, expenses, etc.).
- Do not mix payment-ready cycle time with paid cycle time without labeling.
- Label sample data ILLUSTRATIVE in every worksheet.
- Cite Ardent figures only as external context — never as Evidence Room targets.

12. Implementation phases (0–10)

Phase	Workstream	Exit
0	Foundation — readiness & baseline KPIs	Scorecard + baselines
1	Operating model & exception taxonomy	Published taxonomy
2	A01 Intake shadow → draft	Charter + evidence quality
3	A04 Triage + A03 Match propose	Controls matrix live
4	A10 Duplicate signals	Precision sampling
5	A07 Approval propose (narrow)	SoD tests
6	A05/A06 GR & PO quality	Preventable defect metrics
7	A08/A09 Resolution & follow-up	Template approval
8	Bounded execute for clean match class	Earn-in review
9	A11–A14 close & reporting drafts	Close owner accept
10	Scale, Team packs, Custom edges	Governance board roadmap

13. Integration posture

The OS sits beside your stack. Prefer read APIs and draft artefacts first. Write-backs — if ever — are limited to explicitly approved fields after earn-in, with compensating controls and logging.

- No silent master-data writes
- No payment release authority on agents
- Human identity on every money-adjacent confirmation
- Sandbox parity for prompt/model changes

14. Security & privacy notes

- Minimise PII and bank data sent to external LLM tools
- Prefer enterprise-approved model endpoints with retention controls
- Keep prompt/model change control in the same evidence trail as process changes
- Align retention of evidence packs with your audit and privacy policies

15. Illustrative economics (labels matter)

Business-case worksheets in the Pro pack are labeled ILLUSTRATIVE. Replace every input with measured internal data. Industry averages such as Ardent Partners 2025 (\$9.84 cost/invoice, 18.4% exceptions, 35.4% STP) appear on Notes sheets only — not as embedded targets or promises.

16. Governance board

Stand up a board with AP Manager, Controller (or delegate), Controls, and IT. Meet monthly on KPIs and samples; per release on change tickets; per incident on demotion; quarterly on portfolio roadmap.

17. Fit with Team and Custom

Team adds multi-seat RACI, workshops, and change packs for rollout. Custom designs 1–3 client-specific blueprints when ERP objects or policies diverge from the Pro spine. Fees are fixed-band, not contingent on savings claims.

18. Buyer checklist

- Controller or AP Manager owns the OS — not only IT innovation
- Sandbox available before any write-back
- Internal Audit briefed on earn-up and evidence standards
- Illustrative ROI worksheets replaced with measured inputs
- Licence allows internal use; no public republishing of blueprint text
- Kill-switch tested; demotion examples rehearsed

19. What success looks like in year one

- Baselines published and trusted
- Taxonomy live; exception ageing visible
- At least two agents at Propose with sampled QA
- Zero unsupervised payment releases
- Governance board meeting on cadence with demotion examples on record

20. Anti-patterns

- Prompt libraries without charters
- Shadow agents in personal LLM chats
- Optimism-scored diagnostics
- Guaranteed savings marketing
- Calling A10 'fraud prevention'
- Skipping SoD for speed
- Treating survey averages as SLAs

21. Operating principles — charter discipline

No agent enters Propose without a written charter covering purpose, allowed invoice classes, forbidden actions, evidence standard, KPI gates, owner, and kill-switch. Charters are versioned.

22. Evidence packs

Each material decision leaves a pack: inputs used, rule/model version, confidence or variance table, human override if any, and timestamped identity of the accountable person.

23. Demotion culture

Demotion is not failure theatre — it is control hygiene. Precision slips, SoD near-misses, or evidence gaps trigger automatic return to Draft or Observe until the gate is re-earned.

24. How to use this overview

Share with sponsors to set expectations. Use the full Pro pack for implementation. Pair with Team when people change is the bottleneck. Pair with Custom when your ERP objects refuse to fit generic blueprints.

25. Claims blacklist (never use in customer materials)

- Guaranteed cost/invoice reduction to any industry average
- Eliminate fraud / guaranteed fraud detection
- Audit pass guaranteed / SOX certified by Evidence Room
- Autonomous payment release without human identity

Disclaimer

Evidence Room provides methodologies, diagnostics, and design artefacts. We do not guarantee cost reduction, ROI, fraud prevention, or audit outcomes. Industry benchmarks attributed to Ardent Partners (2025) — average cost per invoice **\$9.84**, exception rate **18.4%**, STP **35.4%** — are third-party reference points only and are not Evidence Room performance claims. Agents must operate under your controls, policies, and human accountability. Evidence Room is not your ERP vendor, bank, or auditor.

AI agents can be wrong. Humans remain accountable for approvals, postings, payments, supplier master changes, and representations to auditors and regulators.

Appendix — Evidence pack checklist (1)

Every material agent decision should leave an evidence pack an auditor can follow.

This appendix reinforces Evidence Room operating discipline. It is educational and does not modify your licence, create warranties, or guarantee operational outcomes.

- Inputs used and source system references
- Rule or model version hash
- Confidence or variance table
- Human override identity and rationale
- Timestamp and Invoice Case ID
- Kill-switch status at time of action

Remember: Observe → Recommend → Draft → Execute (narrow) → Expand. Demotion is hygiene. Ardent Partners (2025) benchmarks are third-party survey averages for context only.

Operating note 1: maintain named owners for agent, control, and KPI. Refuse unsupervised payment release. Keep evidence packs complete. Review overrides weekly. Treat silence in logging as a defect.

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Appendix — Kill-switch protocol (2)

A kill-switch is useless if nobody has rehearsed it.

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- Named operators
- Disable write paths first
- Notify AP Manager / Controls / IT
- Record demotion stage
- Preserve logs
- Retest after prompt/model change

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Appendix — Sample QA method (3)

Sampling beats vibes. Define sample size and demotion trigger before Propose.

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- Stratified samples by invoice class
- Score TP/FP/TN/FN
- Publish weekly precision
- Demote on consecutive misses
- Label worksheets ILLUSTRATIVE until measured
- Never substitute Ardent averages for your pass bar

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Appendix — SoD reminders (4)

Speed is not a reason to collapse segregation of duties.

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- Vendor setup ≠ entry ≠ payment release
- Bank changes dual control
- Agents never self-approve
- Force-pay limited and logged
- Access reviews annually
- Document compensating controls

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Appendix — Claims discipline (5)

Keep customer-facing claims bounded.

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- No guaranteed cost/invoice reduction
- No fraud elimination claims
- No audit-pass guarantees
- Ardent 2025 figures are external context only (\$9.84 / 18.4% / 35.4% STP)
- Label models ILLUSTRATIVE
- Humans remain accountable

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Appendix — Pilot retrospective (6)

Close every pilot with a structured retrospective before expansion.

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- KPI movement vs baseline
- Control KPIs that must not worsen
- Evidence pack quality sample
- False positive/negative themes
- Demotion events
- Go / no-go / demote decision

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Appendix — Exception taxonomy starter codes (7)

Stable codes beat free-text chaos.

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- EX-PRICE
- EX-QTY
- EX-PO
- EX-GR
- EX-TAX
- EX-DUP
- EX-BANK
- EX-APPR
- EX-OTHER

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Appendix — Stage promotion record (8)

Promotion without a record is not governance.

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- Agent ID and charter version
- From-stage → to-stage
- Sample metrics
- Owner sign-off
- Controller acknowledgment for Execute
- Demotion criteria restated
- Effective and review dates

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